MERIDIAN

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The MERIDIAN Program: An Empirical Evaluation of a Systematic Multi-Asset Trading System

Out-of-sample and live evidence from Bitcoin, Ether, gold and silver, 2025 – 2026

Musab Muhammad
Meridian · Correspondence: musabmuhammad1470@gmail.com

ABSTRACT

MERIDIAN is a fully systematic trading program that applies one proprietary rule set, with asset-class-specific parameters, to Bitcoin, Ether, gold and silver contracts for difference (CFDs). We test whether its edge survives recorded trading costs and data that played no part in building it, replaying five-minute price data from the broker Exness, through 7 October 2026, in one shared account that risks 1% of the balance per trade under a 3.5% daily loss limit. On the development sample (January–September 2026), on which the current settings were chosen, the program closed 716 trades with a mean result of +0.22R (t = 3.70) and gained 339.9% with a 13.5% maximum drawdown. Run once with every setting frozen on May–December 2025 data obtained afterwards, it gained 21.3% over 258 trades (profit factor 1.11; mean +0.09R, t = 0.82), with long and short trades contributing almost equally, a 20.2% drawdown and no statistical significance on its own; one account over both periods gained 399.0%. The edge appears to belong to the rule family rather than to the chosen settings: 807 of 1,000 variations were profitable in both periods, but 2026 rankings did not predict 2025 rankings. The rules were profitable on litecoin, never used in development, and unprofitable on two currency pairs. Five of the seven visible trades in the tested markets on a live account supplied by the author, which ran an earlier version, match its backtest trades, with prices within 0.03%. The development-sample figures are therefore an upper bound: the evidence points to a positive but considerably smaller edge, which a forward record of the current version must confirm.

Keywords: systematic trading · backtesting · out-of-sample validation · crypto-assets · precious metals · risk management · backtest overfitting

+339.9%	+21.3%	+399.0%	20.2%	807	5 of 7
Development sample in-sample, 2026	**Out-of-sample test** May – Dec 2025, frozen	**Full period** one account, to Oct 2026	**Maximum drawdown** full period, MTM	**Of 1,000 variations** profitable in both periods	**Live trades matched** by the backtest

PRINCIPAL FINDINGS

- **Positive out of sample, but smaller.** With all settings frozen, the program gained +21.3% on May–December 2025 data (258 trades, profit factor 1.11), from long and short trades alike; the mean result (+0.09R, t = 0.82) is not significant on its own.
- **A property of the rule family.** 807 of 1,000 rule variations were profitable in both periods, and litecoin, never used in development, gained +37.0%; currency pairs lost money.
- **Backtest consistent with live fills.** Five of seven visible live trades in the tested markets coincide with trades in the backtest of the version then running (5.7); four share an entry or exit price to the cent.
- **Material risk; development results optimistic.** The full-period drawdown was 20.2% at 1% risk per trade. Version 6.2 ranked 3rd of 1,000 variations on 2026 data but 421st on 2025, so its development-sample return overstates what to expect.

Except for the author-supplied account record in Section 5.3, all results are hypothetical backtest results. Simulated performance is not a reliable indicator of future results. Limitations are set out in Section 8 and disclosures in Appendix B.

1. Research objective and hypotheses

Short-horizon trading in crypto-assets and precious metals is dominated by noise, shifting volatility regimes and transaction costs, and backtests refined on the data they report tend to overstate performance (Bailey et al., 2014; Harvey & Liu, 2015). This study therefore asks not only whether the MERIDIAN rule set was profitable on the data used to develop it, but whether that profitability survives data, markets and live executions that played no part in its development. We evaluate three hypotheses:

- **H1 (edge).** The mean trade result, measured in units of risk (R), is greater than zero after recorded spreads.
- **H2 (consistency).** Profitability is not confined to a single sub-period, market or trade direction.
- **H3 (out-of-sample).** With all rules and settings frozen, the program remains profitable on data and markets not used in its development, and the backtest reproduces live executions.

H1 and H2 are first tested on the development sample (Section 4), where the results are in-sample, and H3 in Section 5. Account-level risk (drawdown, volatility and exposure) is assessed descriptively in Section 6.

2. Methodology and quantitative framework

2.1 System architecture

MERIDIAN separates signal generation from risk control (Figure 1). A proprietary signal engine identifies a specific price-structure condition, using one parameter profile for crypto-assets and another for precious metals; the two profiles differ in parameter values. Each candidate trade must then pass three context filters, any of which can veto it: a directional-context filter, a volatility-regime filter and an execution-cost screen. Signal rules, filter definitions and parameter values are proprietary and are not disclosed. The rules have also been coded as a MetaTrader 5 Expert Advisor, whose live trades are compared with the backtest in Section 5.3.

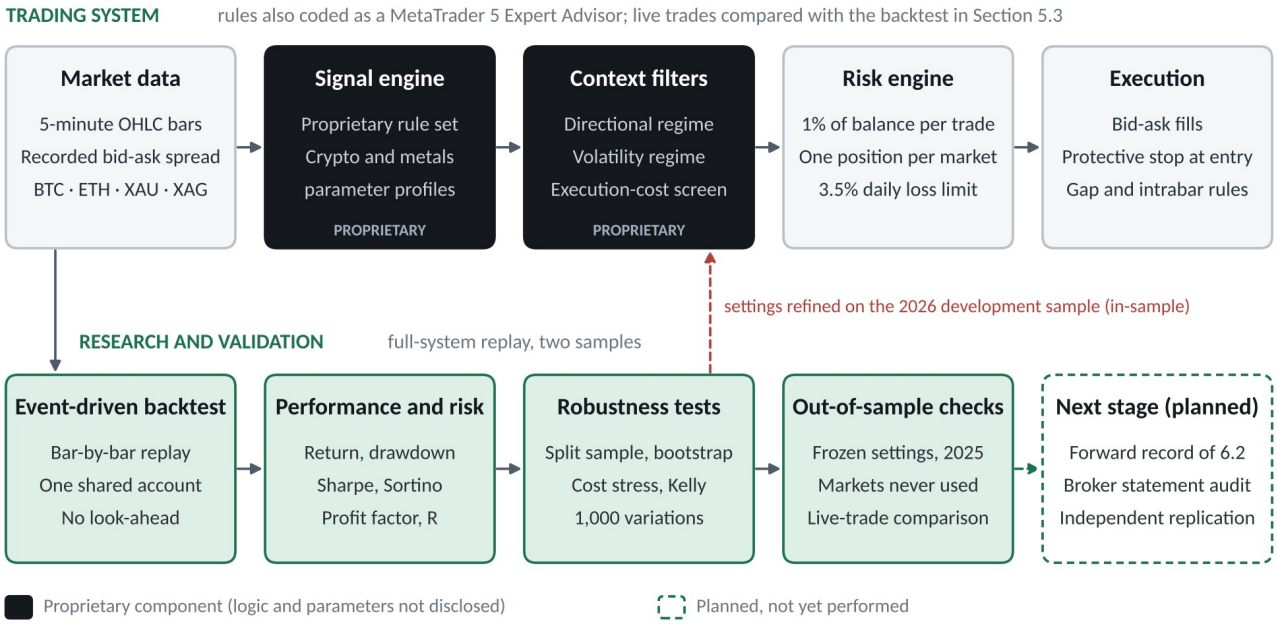


Figure 1. Research and system architecture. Dark components are proprietary and shown only by function. The dashed red arrow marks the refinement of settings on the 2026 development sample (Section 6.7); the out-of-sample checks were added in this edition, and the dashed green box marks steps not yet performed.

2.2 Risk and portfolio rules

The risk rules are disclosed in full because they determine the risk profile:

- **Fixed-fractional sizing.** Each position is sized so that a fill at its protective stop loses 1% of the realized account balance; gains and losses compound.
- **One position per market.** No more than four positions can be open at once.
- **Daily loss limit.** New trades are blocked whenever the day's realized losses plus the open risk of all positions at their stops could exceed 3.5% of the day's opening balance. Price gaps can carry losses beyond the limit.
- **Protective stops.** Every order carries a stop from the moment of entry.

2.3 Backtest engine and execution model

Trades are replayed bar by bar, in chronological order, across all four markets in one shared account, so positions overlap in time, compete for the daily loss limit and are sized from the same balance, as they would be in live operation. Buys fill at the ask and sells at the bid, using the spread recorded for the entry bar throughout the trade; recorded bar spreads can understate execution spreads. When a bar spans both the stop and another exit level, the stop is assumed to be hit first (a conservative convention), and a price gap through any level fills at the bar's opening price. Signal and filter inputs use only information available at the moment of entry (completed bars and the prevailing spread), so no future price information enters any decision. In the base case the only trading cost is the recorded spread; commissions and slippage are stress-tested in Sections 6.5 and 7. Overnight financing is not modelled (7–13% of trades were held through a daily rollover), position sizes are not rounded to lot increments, and margin is not modelled. The engine is deterministic: the same data reproduce every backtest figure in this paper exactly.

2.4 Evaluation framework

Performance is measured on mark-to-market (MTM) equity, which values open positions at every five-minute close, and on the closed-trade balance. We report returns, drawdowns, daily and monthly return statistics, hit rate, profit factor, the mean trade result in R with a bootstrap confidence interval, annualized Sharpe (Sharpe, 1994) and Sortino (Sortino & Price, 1994) ratios of daily MTM returns, and exposure. Robustness is examined with a split-sample comparison, a day-block bootstrap (Efron & Tibshirani, 1993), cost stress tests and a Kelly (1956) analysis of position sizing; out-of-sample evidence is described in Section 3.2. Appendix A defines each measure.

3. Data and testing design

3.1 Data

The study uses five-minute OHLC bars with a per-bar recorded spread for four CFDs (Bitcoin, BTCUSD; Ether, ETHUSD; gold, XAUUSD; silver, XAGUSD), exported by the author from the MetaTrader 5 server of the broker Exness. The files' clock is UTC (gold's daily trading break falls at 21:00 in summer and 22:00 in winter), consistent with Exness's stated server time of GMT+0 (Exness, n.d.), and every closing price visible on the author's live account (Section 5.3) lies inside the corresponding five-minute bar, so the files and that account share one price feed. The 2026 files run from 1 January to 7 October 2026, days before this paper was completed; the 2025 files begin in mid-May (metals) or late October (crypto-assets) and end on 30 December 2025. Timestamps were converted to a single reference time zone, which also defines the trading day. Recorded spreads of zero indicate missing values and were replaced with the market's median recorded spread (44.0% of 2026 Ether bars; the 2025 Ether file has no spread record and uses Ether's 2026 median as a share of price). Table 1 summarizes the development-sample data.

Table 1. Data and market conditions, development sample (4 January – 30 September 2026)

Market	Five-minute bars	Median spread (% of price)	Price change	Annualized volatility	Largest decline	Positive months
BTCUSD	77,523	0.006%	−9.6%	43.1%	38.6%	5 of 9
ETHUSD	77,523	0.006%	−14.9%	58.9%	53.4%	5 of 9
XAUUSD	52,837	0.002%	−5.2%	29.3%	28.6%	4 of 9
XAGUSD	52,770	0.026%	−19.0%	60.9%	53.2%	5 of 9
Equal-weight basket	–	–	−12.2%	–	–	3 of 9

Price change from the 4 January to the 30 September daily close. Volatility from daily close-to-close returns (crypto-assets: all days, ×√365; metals: weekdays, ×√252). Largest decline: maximum peak-to-trough fall in daily closes. Equal-weight basket: equal amounts bought in each market at the 4 January close and held without rebalancing.

3.2 Testing design

The evaluation has four parts, summarized in Table 4 (Section 5).

- **Development sample**, 4 January – 30 September 2026: the year-to-date window on which version 6.2's settings were chosen (Section 4). The volatility filter's inputs were not yet available for 15 of the first 16 trades, which were allowed by default (+8.0R in total); one position open at the cut-off is excluded.
- **Out-of-sample test**, 20 May – 30 December 2025: the frozen configuration run once on data that the author supplied only after version 6.2 and the first edition of this paper had been completed; no setting was changed afterwards. The crypto-asset files begin in late October, so crypto-assets contribute only November and December.
- **Full period**: one continuous account from 20 May 2025 to 7 October 2026.
- **Unseen markets**: litecoin and two currency pairs (EURUSD, GBPUSD), never used in development, run with the crypto-asset profile. Litecoin has no usable spread record and is run at zero spread, with a stress test at Ether's

typical spread; the currency pairs use recorded spreads (mostly zero, as on a raw-pricing account) plus $7 per lot commission.

A live account record supplied by the author is then compared trade by trade with the backtest. Every result comes from May 2025 onwards, so the evidence reflects current spreads, liquidity and volatility. The development sample was also demanding: all four markets fell, by 5.2% (gold) to 19.0% (silver), and an equal-weight buy-and-hold basket lost 12.2% and rose in only three of nine months (Table 1, Figure 3).

4. Development-sample results

This section reports the development sample in detail. Because version 6.2's settings were chosen on these data, the figures are in-sample and should be read as an upper bound on future performance; Section 5 reports the out-of-sample evidence. Table 2 summarizes performance, and Figure 2 shows the equity and drawdown paths.

Table 2. Performance summary, development sample, 4 January – 30 September 2026 (1% risk per trade, 3.5% daily loss limit, recorded spreads)

Returns and risk		Trades and exposure	
Net return (9 months, not annualized)	+339.9%	Closed trades	716
Maximum drawdown, MTM equity	13.5%	Hit rate	38.7%
Maximum drawdown, closed balance	12.3%	Profit factor	1.39
Lowest equity vs. starting capital	−1.0%	Mean trade result	+0.22R
Positive months	9 of 9	95% bootstrap interval, mean	+0.11R to +0.34R
Best / worst month	+48.6% / +1.0%	t-statistic (two-sided p)	3.70 (0.0002)
Positive weeks	30 of 40	Longest losing streak	11 trades
Annualized volatility (daily MTM)	50.3%	Largest single-trade loss	−1.9% of equity
Sharpe ratio, annualized (s.e.)	4.2 (1.2)	Time with a position open	41.7%
Sortino ratio, annualized	8.4	Gross exposure, median / max	1.6× / 18.2×
Net return ÷ maximum drawdown	25.3	Correlation with each market	0.01 to 0.03

R = amount risked per trade. Sharpe and Sortino ratios from 270 daily MTM returns, zero risk-free rate, ×√365; standard error after Lo (2002). Monthly returns on the closed-trade balance; on an MTM basis all nine months are also positive. Gross exposure is the notional value of open positions divided by equity while at least one position is open; correlations use daily returns.



Figure 2. Cumulative return of mark-to-market equity (top) and drawdown from the running equity peak (bottom), development sample, 1% risk per trade. The final value excludes one position still open at the cut-off (unrealized loss of 0.7% of equity).

Equity rose in every month, though not smoothly (Figure 2). The largest drawdown, 13.5% on an MTM basis, ran from 13 April to 23 April and was recovered by 11 May, and five separate drawdowns exceeded 10%. Monthly returns ranged from +1.0% in May to +48.6% in July (Figure 3). Because sizing compounds, later months contributed more in dollar terms; the account more than doubled between the end of June and the end of September.

All four markets and both trade directions were profitable (Table 3). Bitcoin and gold generated 75% of net profit; silver contributed 4.9% from 51 trades. Long and short trades had near-identical profit factors (1.37 and 1.40), and daily returns were essentially uncorrelated with each underlying market (0.01 to 0.03) and with the equal-weight basket (0.03), consistent with returns arising from the trading rules rather than from directional market exposure.

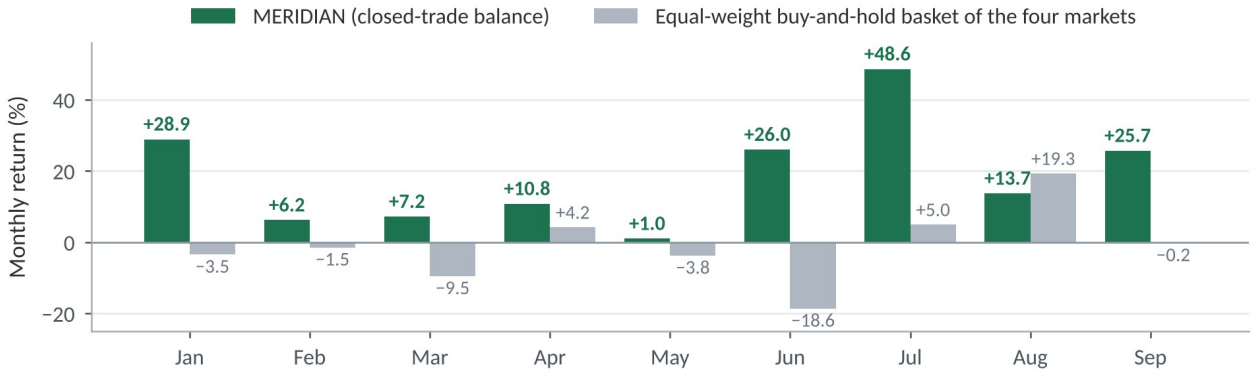


Figure 3. Monthly returns of MERIDIAN (closed-trade balance) and of an equal-weight buy-and-hold basket of the four markets, development sample.

Table 3. Profit attribution by market, asset class and trade direction, development sample

Segment	Trades	Profit factor	Contribution to return (pp)	Share of net profit
Bitcoin	280	1.44	+137.0	40.3%
Ether	268	1.22	+67.2	19.8%
Gold	117	1.69	+119.0	35.0%
Silver	51	1.20	+16.6	4.9%
Crypto-assets	548	1.33	+204.2	60.1%
Precious metals	168	1.53	+135.6	39.9%
Long trades	356	1.37	+161.4	47.5%
Short trades	360	1.40	+178.4	52.5%
All trades	**716**	**1.39**	**+339.9**	**100.0%**

Contribution = segment net profit as a percentage of starting capital; each grouping sums to the total return (differences due to rounding).

5. Out-of-sample and live evidence

5.1 Frozen settings on 2025 data

Table 4 compares the development sample with every test that used data or markets outside it. With all settings frozen, the program gained +21.3% between 20 May and 30 December 2025 (Figure 4a), from long trades (+10.0 percentage points) and short trades (+11.4) alike; gold, silver and Bitcoin were profitable, while Ether lost about as much as Bitcoin gained. Five of eight months were positive (May +3.0%; Jun +0.5%; Jul −3.3%; Aug −3.2%; Sep −2.3%; Oct +7.8%; Nov +16.7%; Dec +1.8%; May is partial). The account spent 97 days below its starting value between mid-June and early November (at worst 9.1% below, on 12 October 2025), and the maximum drawdown, 20.2%, came in December: well above the development sample's 13.5%, but close to the typical drawdown indicated by the development-sample bootstrap (Section 6.4).

Table 4. Development, out-of-sample and full-period results (1% risk per trade, 3.5% daily loss limit)

Test	Period	Return	Max. drawdown	Trades	Profit factor	Mean (t-stat.)
Development (settings chosen)	4 Jan – 30 Sep 2026	+339.9%	13.5%	716	1.39	+0.22R (3.70)
Out-of-sample (settings frozen)	20 May – 30 Dec 2025	+21.3%	20.2%	258	1.11	+0.09R (0.82)
Full period, one account	20 May 2025 – 7 Oct 2026	+399.0%	20.2%	1,006	1.32	+0.17R (3.37)
Litecoin (never used)	23 Sep 2025 – 30 Sep 2026	+37.0%	15.8%	485	1.11	+0.07R (1.21)
EURUSD + GBPUSD (never used)	11 Jun 2025 – 30 Sep 2026	−71.1%	73.7%	813	0.78	−0.14R (−3.17)
without commission		−38.1%	45.4%	820	0.91	−0.05R (−1.09)

Drawdowns are mark-to-market. Mean = mean trade result in R, with its t-statistic against zero. Litecoin at zero spread (with Ether's typical spread: +27.2%); currency pairs with recorded spreads and $7 per lot commission. The 2026 part of the full-period account (1 January – 7 October, 748 trades) returned +311.3% (mean +0.20R, t = 3.49); it differs from the development-sample run because indicators start from 2025 history, trading begins on 1 January and the run ends on 7 October.

The out-of-sample mean, +0.09R per trade (bootstrap 95% interval −0.12R to +0.31R), is about two-fifths of the development-sample figure and is not statistically different from zero on its own (t = 0.82, two-sided p = 0.41). Seven months give little power: even if the true mean were +0.10R, a sample of 258 trades would yield t > 2 only about one time in seven. Over the full period, one continuous account gained +399.0% with a 20.2% maximum drawdown and was positive in 15 of 18 months; started with $5,000, the size of the author's live account, it would have ended at $24,949.

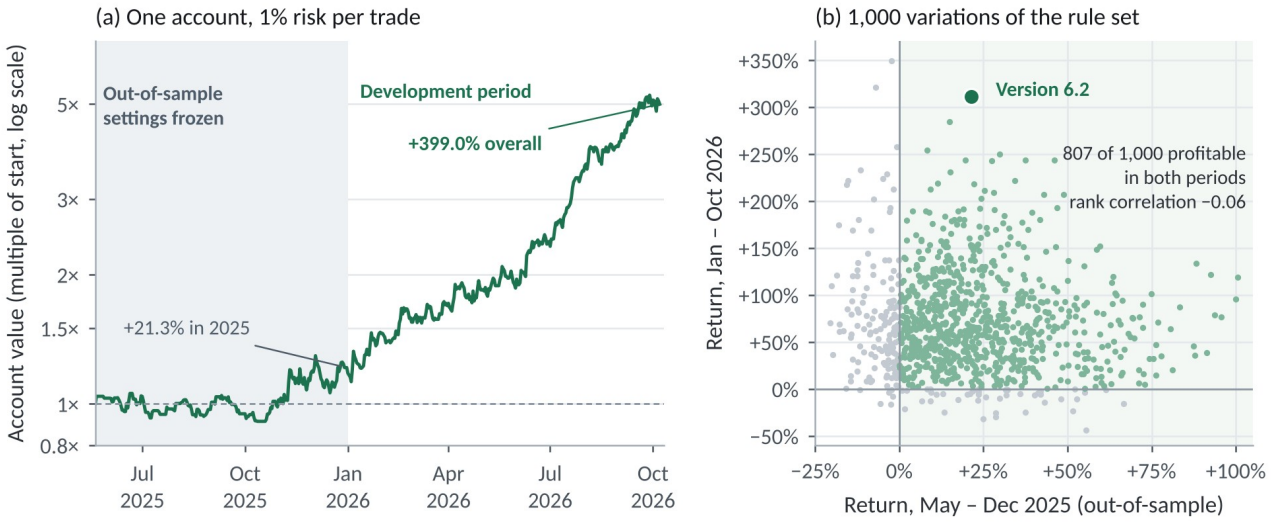


Figure 4. (a) Value of one account at 1% risk per trade, 20 May 2025 – 7 October 2026 (daily mark-to-market, log scale); the shaded period is out-of-sample. (b) Returns of 1,000 variations of the rule set in May–December 2025 (out-of-sample) and January–October 2026; green points were profitable in both periods.

5.2 Rule variations and unseen markets

To see whether the out-of-sample profit depends on the particular settings, we replayed 1,000 variations formed by combining the alternatives for each component that changed between versions 5.7 and 6.2: five directional filters, twenty crypto-asset window and exit combinations, five metals exit rules, and the volatility filter on or off. Of these, 849 were profitable on the 2025 data, 952 on the 2026 data and 807 (80.7%) in both (Figure 4b); the median variation gained +16.7% in 2025. Rankings did not carry over, however: across variations, 2026 and 2025 returns were unrelated (rank correlation −0.06), and version 6.2, 3rd of 1,000 on the 2026 data, ranked 421st on 2025, ahead of 58% of variations. The edge therefore appears to belong to the rule family, while the specific settings selected on the development sample add little that persists, the expected signature of in-sample selection.

The rules transferred to one market outside the development set but not to another asset class. On litecoin the program gained +37.0% over a year (+19.4% in the 2025 months and +14.7% in 2026; +27.2% with Ether's typical spread), with a profit factor of 1.11. On EURUSD and GBPUSD it lost 71.1% after commission (38.1% before), with a significantly negative mean (t = −3.17). MERIDIAN should therefore be treated as a crypto-asset and precious-metals program; its edge does not extend to major currency pairs.

5.3 Live account record

The author supplied a screenshot of the closed-position history of a MetaTrader 5 account which, according to the author, is a $5,000 proprietary-trading-firm account (Figure 5). Its summary shows a deposit of $5,000.00, trading profit of $3,761.60, commission of $346.16, no swap and a balance of $8,415.44: a net gain of +68.3% (+75.2% before commission, which absorbed 9.2% of gross profit). Only the last ten closed positions are visible, from 5 to 14 August 2026, so most of the gain came from earlier trades that are not shown. The screen does not identify the start date, broker, account type or which trades the Expert Advisor placed, and the record has not been audited.

The visible trades were compared with the backtest of version 5.7, the version in use before the research for this paper began (version 6.2 did not yet exist in August). Of the seven visible trades in the four tested markets, five coincide with backtest trades (Figure 5, right): same market, direction and outcome, closing in the same five-minute bar. Four of them share the backtest's entry or exit price to the cent, and all their prices agree within 0.03%. The two remaining gold trades have no backtest counterpart, and the three trades in crude oil and currency pairs are outside the study. In the other direction, the backtest closed 16 further trades in the four markets during the visible period that are absent from the account, so the program was evidently not running on all four markets throughout. The record therefore supports the fidelity of the backtest's simulated entries and exits where both traded, and shows that the strategy family has been traded live. It is not a track record of version 6.2 (only two of the matched trades also occur in that version's backtest), and its profit total cannot be attributed to MERIDIAN without the full broker statement.



Closed (UTC)	Market, side	Result	In v5.7 backtest	Price agreement
5 Aug 20:20	Bitcoin, buy	Profit[1]	Yes	Within 0.03%
6 Aug 10:21	Crude oil, sell	−$78.72	Not tested	–
6 Aug 12:56	Bitcoin, buy	−$81.06	Yes	Entry identical
6 Aug 13:09	Gold, sell	+$135.12	Yes	Entry identical
6 Aug 13:39	Ether, buy	+$163.90	Yes	Exit identical
6 Aug 15:55	Gold, buy	−$74.27	No	–
7 Aug 06:56	Gold, buy	+$96.29	No	–
7 Aug 12:55	Bitcoin, buy	+$115.89	Yes	Entry and exit identical
10 Aug 15:23	GBPUSD, buy	+$64.24	Not tested	–
14 Aug 16:13	EURUSD, buy	+$95.62	Not tested	–

Close times as shown (server time, UTC). Yes: same market, direction and outcome as a version 5.7 backtest trade, closing in the same five-minute bar; prices compared with the backtest's simulated fills. Not tested: market outside this study. [1] Amount not visible on the screen.

Figure 5. Live account record supplied by the author (left: MetaTrader 5 mobile history, cropped to the visible trades and the account summary) and comparison of each visible trade with the version 5.7 backtest (right).

6. Risk and robustness analysis (development sample)

6.1 Volatility, drawdown and exposure

At 1% risk per trade the program is volatile: annualized volatility of daily MTM returns was 50.3%, the worst day −5.3% and the worst week −6.8%. Risk scales almost linearly with the risk setting: at 0.25% per trade the same signals give 12.6% volatility, a 3.5% maximum drawdown and a +47.8% return, with an essentially unchanged Sharpe ratio (4.25 versus 4.24). Gross notional exposure while positioned had a median of 1.6× equity and a maximum of 18.2×, and 55% of crypto-asset trades needed more than 2× equity at entry, so the program could not be run as tested under retail leverage caps such as the 2:1 limit for crypto-asset CFDs introduced by ESMA (2018). The largest single-trade loss was 1.9% of equity, when price gapped through a stop.

6.2 Statistical significance

The development-sample mean of +0.22R has a t-statistic of 3.70 (two-sided p = 0.0002) and a bootstrap 95% interval of +0.11R to +0.34R. This exceeds the threshold of 3.0 that Harvey, Liu and Zhu (2016) suggest for new findings, but a Bonferroni correction for the 577,920 exploratory backtests would require t > 5.35 (t > 4.63 for the 13,440 parameter settings alone); because the configurations are strongly correlated the appropriate threshold lies in between, so the in-sample result cannot be called significant after adjusting for the search. The out-of-sample test needs no such adjustment, and its mean is positive but not significant (t = 0.82). The annualized Sharpe ratio of 4.2 (standard error about 1.2; Lo, 2002) should likewise be expected to fall out of sample (Bailey & López de Prado, 2014); over the full period it was 2.7.

6.3 Consistency across sub-periods

Run separately from the same starting capital, the first half of the development sample (January–June, 466 trades) returned +107.1% with a 13.5% maximum drawdown, and the second (July–September, 250 trades) +107.6% with a 10.8% drawdown (Table 5). Both halves were used in development and the final filter was chosen partly because it improved both, so this check supports H2 in-sample only; the 2025 test in Section 5 is its out-of-sample counterpart.

6.4 Resampling

To test dependence on the particular sequence of events, we generated 2,000 alternative histories by resampling calendar days with replacement (keeping each day's trades together) and re-ran the full account simulation with all risk rules (Figure 6b). The median outcome was +229.8% (5th percentile +67.0%, 95th +577.1%), and 3 of 2,000 histories ended below starting capital. The median closed-balance drawdown was 20.0% and the 95th percentile 32.2%, against 12.3% realized, so a drawdown of about 20% should be regarded as typical, as the out-of-sample drawdown (18.9% closed-balance, 20.2% MTM) later confirmed. Because the histories reuse in-sample trades, they measure sensitivity to the sequence of events, not the probability of a future loss.

Table 5. Robustness checks, development sample (1% risk per trade)

Test	Return	Max. drawdown	Trades	Comment
Full development sample	+339.9%	13.5%	716	Profit factor 1.39
First half (Jan – Jun)	+107.1%	13.5%	466	Profit factor 1.26
Second half (Jul – Sep)	+107.6%	10.8%	250	Profit factor 1.50
Bootstrap, median history	+229.8%	20.0% †	–	2,000 resampled histories
Bootstrap, adverse case	+67.0% (5th pct.)	32.2% † (95th pct.)	–	3 of 2,000 lost money
Extra cost of 0.05R per trade	+205.8%	14.8%	714	8 of 9 months positive
Extra cost of 0.10R per trade	+116.9%	18.1%	713	6 of 9 months positive
Extra cost of 0.15R per trade	+54.7%	23.8%	710	6 of 9 months positive
Break-even extra cost	0.0%	–	–	About 0.21R per trade

Drawdowns are mark-to-market except † (closed-trade balance). Each half starts from the same capital. Extra cost is deducted from the result of every trade.

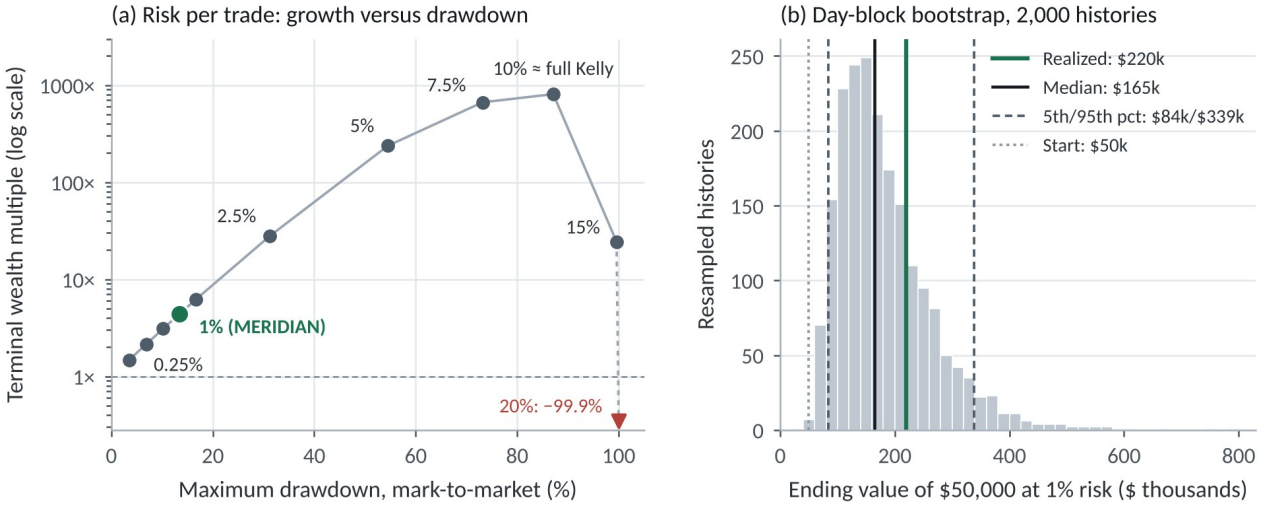


Figure 6. (a) Terminal wealth multiple against maximum MTM drawdown when the development-sample signals are replayed at different risk-per-trade settings, with the daily loss limit at about 3.5 times the risk setting. The backtest has no margin model; higher settings need proportionally larger exposures and are shown only to locate the growth-optimal point. (b) Ending value of $50,000 at 1% risk across 2,000 day-block bootstrap histories (all histories shown).

6.5 Trading costs

Recorded spreads cost a median of 0.013R per trade, so the smallest stress test, 0.05R, is about 3.8 times the recorded cost. An extra 0.05R per trade lowers the development-sample return to +205.8%, 0.10R to +116.9% and 0.15R to +54.7%, and the return reaches zero at about 0.21R per trade. That margin is close to the entire development-sample mean but more than twice the out-of-sample mean of +0.09R, so costs outside the model matter: on the author's live account, commission absorbed 9.2% of gross profit. Commission scenarios are shown in Section 7.

6.6 Position sizing

The growth-optimal (full Kelly) risk per trade estimated from the development-sample trades is 10.0% (bootstrap 5th–95th percentile: 5.3%–14.5%), so the 1% setting is about one-tenth of full Kelly, and a larger fraction of the level implied by the smaller out-of-sample edge. Terminal wealth peaks near full Kelly but with an 87% drawdown, and the 20% setting loses 99.9% of capital despite the positive edge (Figure 6a); fractional Kelly sizing trades growth for drawdown control and robustness to estimation error (MacLean, Thorp & Ziemba, 2011). MERIDIAN's higher-risk configuration (5% per trade, 20% daily limit) recorded a 56.0% maximum drawdown and a −12.5% month over the development sample.

6.7 Development history and selection risk

Version 6.2 is the product of iterative refinement on the development sample. Over the same period, version 5.7 (in use before this research began, and the version whose trades match the live account) returned +94.3% with a 25.9% maximum drawdown, and version 6.1, without the volatility filter, returned +206.9% with an 18.9% drawdown and only six of nine months positive. That filter was selected from 41 candidates, partly because it improved both halves of the sample. Earlier, an exploratory grid of 577,920 backtests (13,440 parameter settings, each with up to nine rule variations) was run on five markets using all data then available (early January to 7 October 2026); 28.2% were profitable in both sub-periods. The development sample's month-by-month consistency is therefore partly a product of selection, as Section 5.2 confirms.

7. Capital illustration: $50,000 starting capital

Table 6 applies the development-sample trade sequence to a $50,000 starting balance under alternative cost assumptions. The figures are hypothetical, not projections, and inherit the development sample's in-sample optimism (Section 5.2). Risk is 1% of the balance per trade with full reinvestment; margin and overnight financing are not modelled. Results scale linearly with capital, so a $5,000 account shows one-tenth of each value (Scenario A: $21,993), apart from lot-size rounding. Scenario B's commission ($3.50 per side per $100,000 of notional, or 0.0035%) is common for raw-pricing FX accounts; crypto-asset and metals schedules can be considerably higher (Scenario C).

Table 6. Hypothetical $50,000 account, development sample (4 January – 30 September 2026)

Scenario	Cost assumption	Ending value	Return	Max. drawdown
A. Backtest as tested	Recorded spreads	$219,933	+339.9%	13.5%
B. Zero-spread account	Commission 0.0035% per side	$212,396	+324.8%	14.3%
C. B with higher commission	Commission 0.01% per side	$149,762	+199.5%	15.4%
D. A with execution slippage	Extra 0.10R per trade	$108,434	+116.9%	18.1%
E. A without compounding	Fixed $500 risk per trade	$127,006	+154.0%	10.6% †
F. A, resampled: median	Recorded spreads	$164,892	+229.8%	20.0% †
5th / 95th percentile		$83,512 / $338,537	+67.0% / +577.1%	

Drawdowns are mark-to-market except † (closed-trade balance). Scenario A month-end values: Jan $64,433; Feb $68,427; Mar $73,383; Apr $81,333; May $82,175; Jun $103,550; Jul $153,920; Aug $175,000; Sep $219,933. At 0.025% commission per side the ending value would be $61,164 (+22.3%) with a 28.9% drawdown.

In the zero-spread commission account (Scenario B), commissions total $21,245; removing the spread also lets more trades pass the execution-cost screen (790 rather than 716). Without compounding, a fixed $500 risk per trade ends at $127,006, so compounding accounts for about 55% of Scenario A's gain. For proprietary-trading-firm accounts, the 3.5% daily loss limit sits inside the daily limits such firms commonly impose (often 5%), but overall limits of about 10% are also common: in the 2025 test the account fell 9.1% below its starting balance, close to such a limit, and peak-to-trough drawdowns of 13.5% to 20.2% would breach a 10% limit measured from the running peak. A lower risk setting gives more room (at 0.25% per trade the development-sample drawdown was 3.5%).

8. Limitations and conclusion

- **Selection on the development sample.** Version 6.2's settings were chosen on January–September 2026 data after an extensive search, and the settings that ranked best there performed like a typical variation out of sample (Section 5.2). The development-sample results are therefore an optimistic estimate of future performance (Bailey et al., 2014; Harvey & Liu, 2015).
- **Short out-of-sample record.** The frozen test covers seven months and 258 trades, with crypto-assets present only from November, and its mean is not statistically significant on its own.
- **Execution model.** Five-minute bars cannot resolve the order of events within a bar, and recorded spreads may understate execution costs; slippage, commissions, financing, partial fills and latency are excluded from the base case.
- **Live record.** The account record is a partial, unaudited screenshot of an earlier version: it shows ten trades, does not identify the broker, account type, full history or which trades the Expert Advisor placed, and includes markets outside this study. Version 6.2 has no live record yet.
- **Data, scope and implementation.** Results rely on one broker's CFD price feed, and the litecoin data lack a spread record. The edge did not extend to currency pairs, and at full size more than half of crypto-asset trades needed leverage above 2:1, which retail rules in some jurisdictions do not permit.

The evidence supports H1 and H2 on the development sample, although the in-sample t-statistic cannot be called significant after adjusting for the search. H3 receives partial support: with all settings frozen, the program was profitable on later-obtained 2025 data, in both directions, and on litecoin; most variations of the rules were profitable in both periods; and the backtest's simulated fills agree closely with live executions where both traded. But the out-of-sample edge per trade is about two-fifths of the in-sample estimate and not yet statistically significant, and the program failed on currency pairs. MERIDIAN is therefore best described as a rule family with a positive but modest edge in crypto-assets and precious metals, with drawdowns of about 20% at 1% risk per trade. The next steps are to keep version 6.2 frozen from the date of this paper, to report its forward record with the metrics used here and the broker's full statements, to replace modelled costs with actual commissions and financing, and to invite independent replication under a confidentiality agreement.

References

Bailey, D. H., Borwein, J. M., López de Prado, M., & Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, *61*(5), 458–471.

Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, *40*(5), 94–107.

Efron, B., & Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.

ESMA (European Securities and Markets Authority) (2018). Decision (EU) 2018/796 of 22 May 2018 to temporarily restrict contracts for differences in the Union. *Official Journal of the European Union*, L 136, 1 June 2018.

Exness (n.d.). *What is commodity trading*. Retrieved 11 October 2026 from https://www.exness.com/commodities/what-is-commodity-trading/

Harvey, C. R., & Liu, Y. (2015). Backtesting. *Journal of Portfolio Management*, *42*(1), 13–28.

Harvey, C. R., Liu, Y., & Zhu, H. (2016). … and the cross-section of expected returns. *Review of Financial Studies*, *29*(1), 5–68.

Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, *35*(4), 917–926.

Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, *58*(4), 36–52.

MacLean, L. C., Thorp, E. O., & Ziemba, W. T. (Eds.) (2011). *The Kelly Capital Growth Investment Criterion: Theory and Practice*. World Scientific.

Sharpe, W. F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, *21*(1), 49–58.

Sortino, F. A., & Price, L. N. (1994). Performance measurement in a downside risk framework. *Journal of Investing*, *3*(3), 59–64.

Appendix A. Definitions

R: the amount at risk on a trade, i.e. the loss if the protective stop is filled at its level. A result of +1R gains the amount risked.

Hit rate: share of closed trades with a positive result. **Profit factor:** gross profits divided by gross losses.

Mean trade result: average result in R (expectancy). Its interval is a percentile bootstrap over trades (10,000 resamples).

MTM equity: balance plus the value of open positions at each five-minute close. **Maximum drawdown:** largest percentage decline from a running peak.

Sharpe ratio: mean divided by standard deviation of daily MTM returns, ×√365, zero risk-free rate. **Sortino ratio:** mean divided by downside deviation, ×√365.

Kelly fraction: the risk per trade that maximizes mean logarithmic growth over the empirical distribution of trade results.

Out-of-sample: data or markets not used at any stage to choose rules or settings. **Rank correlation:** Spearman correlation between the variations' returns in the two periods.

Appendix B. Important disclosures

This paper presents hypothetical (simulated) performance from a backtest, together with an account record supplied by the author. Hypothetical results have inherent limitations: they are prepared with the benefit of hindsight, do not represent actual trading and may not reflect market factors such as liquidity, slippage and execution quality. The account record is reproduced as supplied and has not been audited or verified against broker statements. Past performance, simulated or actual, is not a reliable indicator of future results, and no representation is made that any account will or is likely to achieve results similar to those shown. CFDs are leveraged products and carry a high risk of rapid loss. This document is for research and information purposes only and is not investment advice, or an offer or solicitation to buy or sell any financial instrument. The MERIDIAN trading rules, parameters and source code are proprietary and are not disclosed, and the backtest figures have not been independently verified. Conflict of interest: the author developed the MERIDIAN program and is affiliated with Meridian, which may benefit commercially from its use.